Data Preparation Guide — Seed CSVs

This document provides column-by-column guidance for the two seed data CSVs required before the MVP frontend can be demonstrated. Both CSVs are loaded by management commands during Gate 06.

Execution order: `load_reference_layers` (shapefiles) -> `seed_species` (species CSV) -> `seed_localities` (localities CSV) -> `generate_map_layers`

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1. Species Seed CSV

File: `data/seed/madagascar_freshwater_fish_seed.csv` **Loaded by:** `python manage.py seed_species --csv data/seed/madagascar_freshwater_fish_seed.csv` **Idempotent key:** `scientific_name` (updates existing records on re-run)

Column Reference

#	Column	Required	Type	Allowed Values	Note
1	`scientific_name`	Yes	string	—	Unique identifie undescribed ta: conventional fo `sp. 'epithet'` (e `Bedotia sp. 'ma`
2	`authority`	No	string	—	Taxonomic auth `Holly, 1928`). Le for undescribec
3	`year_described`	No	integer	—	Year of formal c Leave blank for undescribed ta:
4	`family`	Yes	string	—	Family name (e. `Bedotiidae`, `Cich` `Aplocheilidae`)
5	`genus`	Yes	string	—	Genus name
6	`endemic_status`	Yes	enum	`endemic`, `native`, `introduced`	Most Malagasy are `endemic`. Use diadromous/wic species, `introd` invasives.
7	`iucn_status`	No	enum	`EX`, `EW`, `CR`, `EN`, `VU`, `NT`, `LC`, `DD`, `NE`	IUCN Red List c Leave blank if n assessed (treat
8	`iucn_taxon_id`	No	integer	—	IUCN Red List t Enables automa sync. Find at: https://www.iuc

					(search species the URL).
9	`population_trend`	No	enum	`increasing`, `stable`, `decreasing`, `unknown`	IUCN populatio Leave blank if u
10	`cares_status`	No	enum	`CCR`, `CEN`, `CVU`, `CLC`	CARES 2.0 cons priority. Leave b CARES-listed.
11	`taxonomic_status`	Yes	enum	`described`, `undescribed_morphospecies`, `species_complex`, `uncertain`	Default: `describ` `undescribed_mor` for recognized formally undesc
12	`provisional_name`	No	string	—	Informal epithe undescribed ta: `'manombo'`, `'nosi` Leave blank for species.
13	`shoal_priority`	No	boolean	`true`, `false`	Whether this sp appears on the 1,000 Fishes Bl Default: `false`.
14	`fishbase_id`	No	integer	—	FishBase specie at: https://www.fisl
15	`synonyms`	No	string	—	Semicolon-sepa of junior synony former names (`Bedotia longian` imported to dat informational fo curation.
16	`distribution_narrative`	No	string	—	Free text descri geographic ran
17	`habitat_type`	No	string	—	Habitat descrip `streams`, `rivers` `cave systems`).
18	`max_length_cm`	No	decimal	—	Maximum total centimeters.
19	`in_captivity`	No	enum	`Y`, `N`	Whether the sp known to be he captivity. Impor `BooleanField` on model. Useful f display before

					ExSituPopulatic exist. Default: f
20	`captive_institutions`	No	string	—	Comma-separa institution name this species. Int only — not impo database.
21	`notes`	No	string	—	Provenance not curator. Not imp database.

Example Rows

```
scientific_name,authority,year_described,family,genus,endemic_status,iucn_status,iucn_t
Pachypanchax sakaramyi,"Holly,
1928",1928,Aplocheilidae,Pachypanchax,endemic,EN,166478,decreasing,CEN,described,,false
Madagascar; Sakaramy River drainage,streams,8.5,Y,"Cologne Zoo, London Zoo",ZIMS
records confirm holdings
Bedotia sp.
'manombo',,,Bedotiidae,Bedotia,endemic,,,,,undescribed_morphospecies,'manombo',false,,,
Special Reserve area,streams,,Y,"Private breeders (CARES)",CARES 2026 R4.1 priority
Oreochromis niloticus,"(Linnaeus,
1758)",1758,Cichlidae,Oreochromis,introduced,LC,166775,stable,,described,,false,2,,Intr
throughout Madagascar,rivers and lakes,60.0,N,,Major invasive threat to endemic
fauna
```

Guidance for Undescribed Taxa

Including undescribed morphospecies is valuable if:

- The taxon is recognized by specialists (published in literature or known from collections)
- It has conservation significance (threatened habitat, CARES-listed, captive populations exist)
- It has locality data you can provide

If an undescribed taxon has no locality data and no captive populations, it adds limited value to the MVP demonstration.

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2. Species Localities Seed CSV

File: `data/seed/madagascar_freshwater_fish_localities_seed.csv` **Loaded by:** `python manage.py seed_localities --csv data/seed/madagascar_freshwater_fish_localities_seed.csv` **Idempotent key:** `(scientific_name, latitude+longitude, locality_type)` — safe to re-run **Prerequisite:** Species must be loaded first (`seed_species`). Reference layers should be loaded first (`load_reference_layers`) for drainage basin FK assignment.

Master File and Filter Rules

`data/seed/madagascar_freshwater_fish_localities_seed_master.csv` is the unfiltered source (942 rows as of 2026-04-20). The seed CSV above is derived from it by applying these filters:

- **Drop genus-only identifications.** Rows whose `scientific_name` is a single word (e.g., `Bedotia`, `Rheocles`, `Glossogobius`) are excluded. They cannot populate a species profile and have no home in the species-level MVP data model. This removed 233 rows. If a future "indeterminate occurrence" bucket is added, revisit this.
- **Drop out-of-bounds coordinates.** Any row with lat/lng outside the Madagascar extent (lat -26.0 to -11.5, lng 43.0 to 51.0) is excluded as a data-entry error. As of 2026-04-16 this removed 1 row (`Rheocles pellegrini` at lon=42.5, locality "andapa" — inconsistent with Andapa's actual position in northern Madagascar).

Preserve the master file. When re-curating or re-filtering, work from the master and regenerate the seed — do not hand-edit the seed as the source of truth.

Column Reference

#	Column	Required	Type	Default	Allowed Values
1	`scientific_name`	Yes	string	—	—
2	`latitude`	Yes	decimal	—	-26.0 to -11.5
3	`longitude`	Yes	decimal	—	43.0 to 51.0
4	`locality_name`	Yes	string	—	—
5	`locality_type`	Yes	enum	—	`type_locality`, `collection_record`, `literature_record`, `observation`
6	`presence_status`	No	enum	`present`	`present`, `historically_present_extirpated`, `presence_unknown`, `reintroduced`
7	`water_body`	No	string	*(blank)*	—
8	`water_body_type`	No	enum	*(blank)*	`river`, `lake`, `stream`, `cave_system`, `wetland`, `estuary`
9	`coordinate_precision`	No	enum	`exact`	`exact`, `approximate`, `locality_centroid`,

					`water_body_centroid`
10	`year_collected`	No	integer	*(null)*	—
11	`collector`	No	string	*(blank)*	—
12	`source_citation`	Yes	string	—	—
13	`is_sensitive`	No	boolean	`false`	`true`, `false`
14	`notes`	No	string	*(blank)*	—

Auto-Computed Fields (NOT in CSV)

These are set automatically by the management command or model `save()` — do not include them in the CSV:

Field	How It's Set
`drainage_basin`	Spatial query: `Watershed.objects.filter(geometry__contains=point)`. Assigned to the smallest matching watershed by area. Null if no watershed contains the point.
`drainage_basin_name`	Denormalized from `drainage_basin.name` on save.
`location_key`	Computed as `"{lng:.5f},{lat:.5f}"` for uniqueness constraint.
`location_generalized`	For `is_sensitive=true` records: coordinates rounded to 0.1 degree. Public API serves these instead of exact coordinates.

Example Rows

```
scientific_name,latitude,longitude,locality_name,locality_type,presence_status,water_bo
Pachypanchax sakaramyi,-12.5200,49.1800,Sakaramy River near
Joffreville,type_locality,present,Sakaramy River,stream,exact,1928,Holly,Holly
1928,false,Original type locality description
Pachypanchax sakaramyi,-12.5350,49.1650,Sakaramy tributary upstream of
village,collection_record,present,Sakaramy River,stream,approximate,2019,Sparks &
Stiassny,Sparks & Stiassny 2019,false,Recent survey confirmation
Bedotia sp. 'manombo',-23.0800,47.7200,Manombo Special Reserve
stream,observation,present,,stream,exact,2022,Durrell team,Durrell field survey
2022,true,Exact location sensitive — CR species with very restricted range
Ptychochromis insolitus,-14.5200,49.7000,Nosivolo River at
Marolambo,type_locality,historically_present_extirpated,Nosivolo
River,river,exact,1926,Pellegrin,Pellegrin 1926,false,Not observed since original
collection; extirpated from type locality
Oreochromis niloticus,-18.9200,47.5200,Lac Alaotra,observation,present,Lac
```

```
Alaotra,lake,locality_centroid,,,"GBIF occurrence records, multiple
observers",false,Invasive; coordinates placed at lake centroid
```

Guidance: `locality_type`

Value	Use When
`type_locality`	The location where the species was originally described. Each described species should have exactly one.
`collection_record`	A museum voucher specimen or verified collection with coordinates. Highest confidence.
`literature_record`	Coordinates derived from a publication but without a specific voucher. Common for older literature.
`observation`	Field sighting, eDNA detection, or unvouchered record. Includes recent surveys.

Guidance: `coordinate_precision`

This field is critical for data quality assessment on the map.

Value	Use When	Typical Source
`exact`	GPS coordinates from the collector or precise georeferencing	Field surveys, GPS-tagged collections
`approximate`	Coordinates estimated from a described locality but not GPS-verified	Literature with descriptive localities ("near village X")
`locality_centroid`	Point placed at the center of a named locality (town, reserve)	Records with only a locality name, no coordinates
`water_body_centroid`	Point placed at the center of a named water body	Records that say "Lac Alaotra" with no further detail

Rule of thumb: If you're placing a point on a map based on a place name rather than recorded coordinates, use `approximate`, `locality_centroid`, or `water_body_centroid` as appropriate. The map displays all precision levels but the field helps users assess what they're looking at.

Guidance: `is_sensitive`

Set `is_sensitive = true` when:

- The species is CR or EN **and** has a very restricted range (single river, single cave)
- Exact coordinates could enable poaching, collection pressure, or habitat disturbance
- The coordinates are precise enough to locate the actual site

When `is_sensitive = true`, the public API serves coordinates rounded to 0.1 degree (~11km). Tier 3+ users see exact coordinates via Django Admin.

When in doubt, leave as `false`. Most historical collection records and type localities are already published in literature and are not sensitive. Sensitivity is primarily for recently discovered, range-restricted populations.

Guidance: `source_citation`

Be specific enough that a reviewer can trace the record back. Examples:

- `Holly 1928` — original species description
- `Sparks & Stiassny 2003, Syst. Biodiv. 1:313–344` — published revision
- `GBIF.org (accessed 2026-04-15), occurrence ID 12345` — GBIF record
- `Durrell field survey 2022, unpublished` — field data
- `FishBase, accessed 2026-04-15` — online database

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3. Reference Layer Data (Shapefiles)

These are not CSVs — they are shapefiles downloaded from external sources and loaded by management command.

HydroBASINS Watersheds

File: `data/reference/hydrobasins_madagascar_lev06.shp` **Source:** https://www.hydrosheds.org/products/hydrobasins (Africa, Level 06) **Preparation:** Download the Africa level 6 dataset, filter/clip to Madagascar bounding box (lat -26.0 to -11.5, lng 43.0 to 51.0). Export as shapefile. **Expected records:** ~60–100 watershed polygons **Loaded by:** `python manage.py load_reference_layers --watersheds data/reference/hydrobasins_madagascar_lev06.shp`

WDPA Protected Areas

File: `data/reference/wdpa_madagascar.shp` **Source:** https://www.protectedplanet.net/en/thematic-areas/wdpa (monthly download) **Preparation:** Download the current WDPA dataset, filter to `ISO3 = 'MDG'`. Export as shapefile. **Expected records:** ~120–160 protected area polygons **Loaded by:** `python manage.py load_reference_layers --protected-areas data/reference/wdpa_madagascar.shp`

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4. Validation Checklist

Before running the seed commands, verify:

Species CSV

- [ ] All `scientific_name` values are unique
- [ ] All `family` and `genus` values are non-empty
- [ ] `endemic_status` values are one of: `endemic`, `native`, `introduced`
- [ ] `taxonomic_status` values are one of: `described`, `undescribed_morphospecies`, `species_complex`, `uncertain`
- [ ] Undescribed taxa have `authority` and `year_described` blank
- [ ] Undescribed taxa have `provisional_name` set
- [ ] `iucn_status` values (when present) are valid IUCN codes
- [ ] `iucn_taxon_id` values (when present) are valid integers from iucnredlist.org

Localities CSV

- [ ] All `scientific_name` values match a row in the species CSV
- [ ] All coordinates are within Madagascar extent (lat -26.0 to -11.5, lng 43.0 to 51.0)

- [ ] Each described species has at most one `type_locality` record
- [ ] `locality_type` values are one of the four allowed enums
- [ ] `coordinate_precision` reflects how the coordinates were obtained (not always `exact`)
- [ ] `source_citation` is non-empty for every row
- [ ] `is_sensitive` is only set for genuinely sensitive records (CR/EN, restricted range, precise coords)
- [ ] No duplicate rows (same species + same coordinates + same locality_type)

Reference Layers

- [ ] Shapefiles have valid `.shp`, `.shx`, `.dbf`, `.prj` files
- [ ] HydroBASINS features have `HYBAS_ID` and `PFAF_ID` attributes
- [ ] WDPA features have `WDPAID`, `NAME`, `DESIG_ENG`, `IUCN_CAT` attributes
- [ ] Both datasets are in WGS84 (EPSG:4326)

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5. Timeline

Per the conservation map integration spec, the localities CSV is on the critical path for the ECA Workshop demo (June 1–5, 2026). Target completion: **May 25, 2026** (one week buffer).

A "thin seed" fallback (type localities only, ~79 records for described species) provides a meaningful demo if full curation is not complete by that date.